

Uncooled LWIR OEM Thermal Camera Module

BOSON®

NDAA compliant and ITAR free, the Boson longwave infrared (LWIR) OEM thermal camera module set the standard for size, weight, power, and performance (SWaP). Utilizing Teledyne FLIR's advanced image processing and several industry-standard communication interfaces, Boson enables applications from firefighting to unmanned aircraft systems (UAS), security, and automotive development kits, all for as little as 600 mW.

The 12 μm uncooled detector comes in two resolutions – 640 x 512 or 320 x 256 – and multiple frame rate options. Radiometric models offer absolute temperature measurement. With multiple lens configurations also available, Boson offers the widest range of LWIR models from Teledyne FLIR and the most flexibility to integration programs. The easy-to-use Boson SDK, user-friendly GUI, and comprehensive product integration documentation further simplify OEM integrated into higher-level systems.

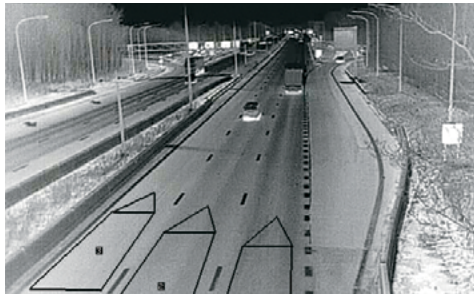
For a plug-and-play sensitivity upgrade, the new Boson+, provides an industry-leading NeDT of <20 mK.



INDUSTRY-LEADING SIZE, WEIGHT, AND POWER (SWAP) WITH RADIOMETRY

A full-featured VGA and QVGA LWIR thermal camera modules starting at 7.5 grams and <4.9 cm³.

- Low power consumption, starting at 600 mW
- 640 and 320 resolutions, 12 μm pixel pitch radiometric LWIR microbolometer
- Rugged construction and stable operation across temperature rating of -40 °C to 80 °C



PROVEN PERFORMANCE AND WIDELY DEPLOYED

Consistent performance, flexibility, and availability of the widest range of LWIR model configurations from Teledyne FLIR.

- High volume manufacturing with off-the-shelf availability
- Accessible third-party accessory kits from numerous third parties
- Adapt performance to use conditions through both hardware and software configurations



DESIGNED FOR INTEGRATORS

Shared mechanical/electrical compatibility across all Boson provides plug-and-play with existing designs.

- Easy-to-use Boson SDK, user-friendly GUI, and comprehensive product integration documentation
- Highly qualified Technical Services team available to support integration
- NDAA compliant and classified under US Department of Commerce jurisdiction as EAR 6A003.b.4.a

SPECIFICATIONS

Thermal Imager	Boson
Resolution	320 x 256 or 640 x 512
Pixel Pitch	12 µm
Spectral Band	Longwave infrared; 8 µm – 14 µm
Sensitivity [NEdT]	<40 mK (Industrial) <50 mK (Professional) <60 mK (Consumer)
Radiometric Temperature Measurement	Available in some models
Frame Rate	60 Hz baseline; 30 Hz runtime selectable, <9 Hz available
Non-uniformity Correction (NUC)	Factory calibrated; updated FFCs with FLIR Silent Shutterless NUC (SSN™)
Solar Protection	Integral
Digital Zoom	1x to 8x zoom
Symbol Overlay	Re-writable each frame; alpha blending for translucent overlay

Lens Options

	320 x 256	640 x 512
Resolution		
Horizontal Field of View (HFOV);	92°; 2.3 mm	95°; 4.9 mm
Effective Focal Length	50°; 4.3 mm	50°; 8.7 mm
	34°; 6.3 mm	50°; 9.2 mm
	24°; 9.1 mm	32°; 13.6 mm
	16°; 14 mm	32°; 14 mm
	12°; 18 mm	24°; 18 mm
	6°; 36 mm	18°; 24 mm
	4°; 55 mm	12°; 36 mm
		8°; 55 mm
		6°; 73 mm

Physical Attributes

Size	21 x 21 x 11 mm (0.83 x 0.83 x 0.43 in) without lens
Weight	7.5 g (0.26 oz) without lens
Precision Mounting Holes	Four tapped M1.6x0.35 (rear cover)

Interfacing

Input Voltage	3.3 VDC
Power Consumption	Varies by configuration; as low as 500 mW
Video Channels	CMOS or USB-2
Control Channels	UART or USB
Configurable GPIO	Up to 11; user configurable

Environmental

Operating Temperature Range	-40 °C to 80 °C (-40 °F to 176 °F)
Non-Operating Temperature Range	-50 °C to 85 °C (-58 °F to 185 °F)
Shock	1,500 g @ 0.4 msec
Operational Altitude	12,192 m (40,000 ft) (max altitude of a commercial airliner or airborne platform)

Specifications are subject to change without notice.
For the most up-to-date specs, go to www.flir.com/boson

SANTA BARBARA
Teledyne FLIR LLC, Inc.
6769 Hollister Ave.
Goleta, CA 93117
PH: +1 805.690.6602

EUROPE
Teledyne FLIR LLC, Inc.
Luxemburgstraat 2
2321 Meer
Belgium
PH: +32 (0) 3665 5106

Equipment described herein is subject to US export regulations and may require a license prior to export. Diversion contrary to US law is prohibited. Imagery for illustration purposes only. Specifications are subject to change without notice. ©2025 Teledyne FLIR LLC, Inc.

Approved for public release. Teledyne FLIR Approved [FLIRGTC-SBA-001]

All rights reserved. Revised 06/19/2025

21-0917-OEM-COR-Boson-R-Datasheet-LTR

For more information visit:
www.flir.com/boson